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PARATHYROID HORMONE CAUSES A TRANSIENT RISE IN INTRACELLULAR IONIZED CALCIUM IN VASCULAR SMOOTH MUSCLE CELLS*

Hiroyuki Kawashima

Department of Cell Biology,
Tsukuba Research Laboratories,
Yamanouchi Pharmaceutical Co., Ltd.,
21 Miyukigaoka, Tsukuba City,
Ibaraki 305, Japan

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The effect of parathyroid hormone on intracellular calcium concentration in vascular smooth muscle cells in culture was studied. Human PTH 1-34 (hPTH (1-34)) caused a transient rise in intracellular calcium in a dose-dependent manner at physiological concentrations. The effect of PTH was mimicked by dibutyryl cyclic AMP and inhibited by a PTH receptor antagonist. The effect of PTH was increased in parallel with extracellular calcium concentration and a sustained response was observed when extracellular calcium was 2 mM or higher. The PTH action was blocked by nisoldipin, a calcium antagonist, but not by ouabain, a Na, K-ATPase inhibitor. These data indicate that PTH increases intracellular calcium through its receptor via opening calcium channels. A possible role of this effect in the regulation of vascular tone is also discussed.

Intracellular ionized calcium ($[Ca^{2+}]_i$) is known as a determinant of vascular tone, and thus determines blood pressure, since increased free calcium binds to calmodulin which, in turn, activates myosin light chain kinase, leading to contraction of smooth muscle (1-3). Accordingly, agents that cause an increase in $[Ca^{2+}]_i$ in vascular smooth muscle cells have been known to induce contraction. In contrast, atrial natriuretic peptide (ANP), which relaxes vascular smooth muscles via cyclic GMP, has recently been shown to inhibit a rise in $[Ca^{2+}]_i$ induced by agonists such as norepinephrine (4) and vasopressin (5, 6) in smooth muscle cells.

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Abbreviations used : PTH, parathyroid hormone ; DMEM, Dulbecco's modified minimum essential medium.

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Parathyroid hormone (PTH) is also known to relax vascular smooth muscle cells presumably via cyclic AMP, thereby decreasing blood pressure (7, 8). The effect of PTH on $[Ca^{2+}]_i$ in vascular smooth muscle, however, is not known. In analogy with the effect of ANP, PTH may decrease $[Ca^{2+}]_i$ in vascular smooth muscle cells. On the contrary, it is known that PTH increases $[Ca^{2+}]_i$ in a variety of tissues including the kidney (9) and bone (10). These data led me to investigate the effect of PTH on the $[Ca^{2+}]_i$ in vascular smooth muscle cells, using Fura-2/AM. The Role of extracellular calcium ($[Ca^{2+}]_o$) in the action of PTH was also studied.

MATERIALS AND METHODS

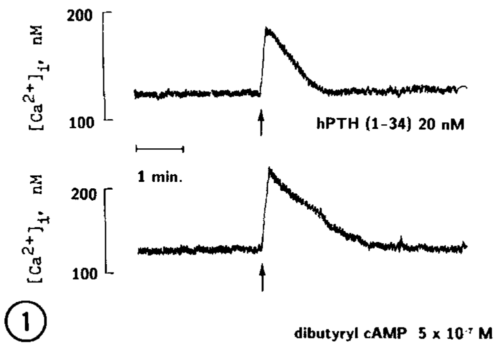
A cell line, A_7r_5, derived from fetal rat aorta, was incubated in Dulbecco's modified Eagle's minimum essential medium (DMEM) supplemented with 10% fetal bovine serum at 37°C in 5% CO_2. At confluence, the cells were treated with 0.25% trypsin-EDTA for 3 min. The solution of trypsin-EDTA was aspirated, and the cells were allowed to stand for 10 min at room temperature. The detached cells were recovered in DMEM with 10% fetal bovine serum, and aliquots of the cell suspension were transferred onto a glass cover slip (13.5 mm ϕ) which was placed in each well of a plate with 24 wells (11). One half milliliter of DMEM supplemented with 5% fetal bovine serum was added to each well, and the cells were incubated until they reached confluence. Cells were washed three times by aspirating and re-adding assay media containing 140 mM NaCl, 5 mM KCl, 10 mM Na_2HPO_4, 20 mM Hepes, 1 mM $CaCl_2$, 5 mM $MgCl_2$, pH 7.4 and incubated with 1 uM Fura-2/AM (Dojindo Laboratories, Kumamoto, Japan) in 0.5 ml of the assay medium for 30 min. Each well received 1.5 ml of the assay medium, and the cells were incubated for an additional 30 min. at 37°C. Then a cover slip was taken out of the well, washed in fresh assay medium and put in a cuvette (10 x 10 mm) containing 1.5 ml assay medium. The cuvette was set in a calcium analyzer CAF-100 (Japan Spectroscopy Co., Ltd., Tokyo, Japan), and fluorescences were measured at 37°C. Emission intensity at 500 nm excited at 340 nm and 380 nm was determined, and the fluorescence ratio F_{340}/F_{380} was obtained. Intracellular ionized calcium concentration was determined according to the method of Tsien et al (12), using the following equation:

$$[Ca^{2+}]_i = K_d(F-F_{min})/(F_{max}-F)$$

in which K_d = 224 nM, F = fluorescence signal, and F_{max} and F_{min} are fluorescences determined in the presence of 20 nM digitonin and 5×10^{-4} M $MnCl_2$, respectively. Each hormone or agent was added to the assay medium in 15 ul of vehicle, respectively. Human synthetic PTH (1-34) and PTH (3-34) were purchased from Peptide Institute, Inc. (Osaka, Japan). Statistical analyses were carried out by Student's t-test, and data were described as a mean ± SEM.

RESULTS AND DISCUSSION

Figure 1 illustrates a typical response of $[Ca^{2+}]_i$ to PTH (20 nM) in vascular smooth muscle cells. Intracellular ionized calcium increased immediately after addition of PTH (within 10 sec) and returned to the basal level in 3 min. The effect of the PTH was dose



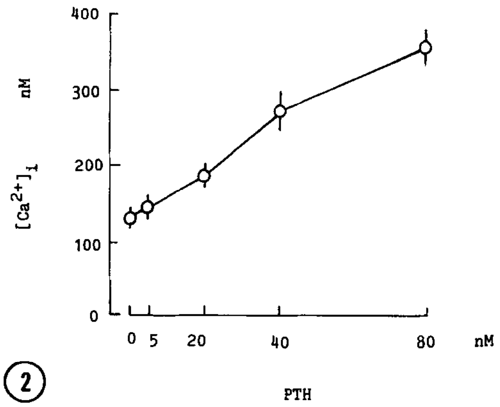


Figure 1. Effects of PTH and dibutyryl cyclic AMP on the intracellular calcium in vascular smooth muscle cells.

Figure 2. Dose-dependent rise in intracellular ionized calcium in response to PTH in vascular smooth muscle cells. Each point and bar represent a mean of 5 incubations ± SEM.

dependent at 20-80 nM (Figure 2). This is similar to the action of PTH on $[Ca^{2+}]_i$ in the kidney and bone cells (9, 10).

Dibutyryl cyclic AMP at 5 x 10^{-7} M mimicked the effect of PTH as shown in Figure 1. This effect of cyclic AMP was also dose-dependent (data not shown). Furthermore, PTH (3-34), a PTH antagonist, inhibited PTH action, and even 200 nM PTH (1-34) failed to appreciably elevate $[Ca^{2+}]_i$ when PTH (3-34) was present (Figure 3). These data indicate that the rise in $[Ca^{2+}]_i$ caused by PTH may be manifested through the PTH receptor via the activation of adenylate cyclase.

To elucidate the mechanism of action of PTH, the effect of extracellular calcium concentration was examined next. As indicated in Figure 4, the effect of PTH was proportional to the extracellular

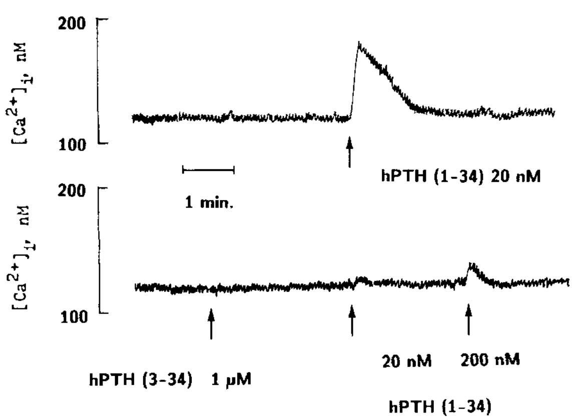


Figure 3. Antagonistic action of PTH (3-34), a PTH antagonist, on the rise of intracellular ionized calcium concentration in vascular smooth muscle cells.

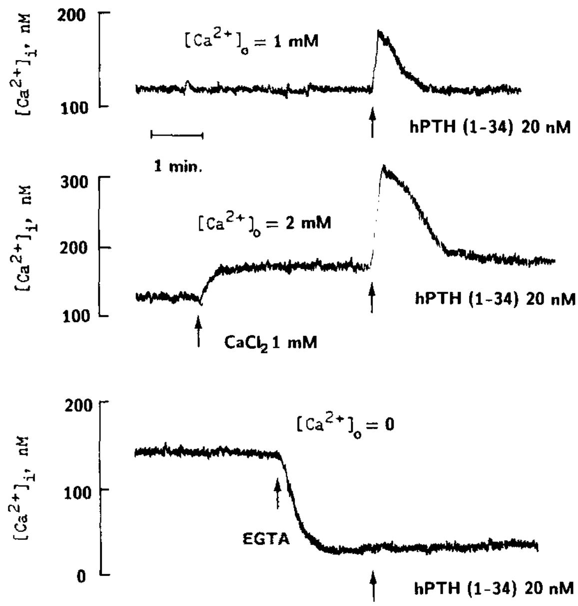


Figure 4. Effects of extracellular calcium concentration on the PTH action in vascular smooth muscle cells. Note that the ionized calcium level does not return to the basal level, but remains slightly higher when $[Ca^{2+}]_i$ = 2mM.

calcium. The basal concentration of intracellular calcium also increased with extracellular calcium. It should be noted that $[Ca^{2+}]_i$ did not return to the basal level, but remained slightly higher at 2 mM extracellular calcium concentration (Figure 4) or higher (Data not shown). When extracellular calcium was removed by adding 2 mM EGTA, the effect of PTH was blocked (Figure 4).

When 10^{-7} M nisoldipin, a calcium channel blocker, was present, the PTH-induced rise in the $[Ca^{2+}]_i$ was completely blocked (Figure 5),

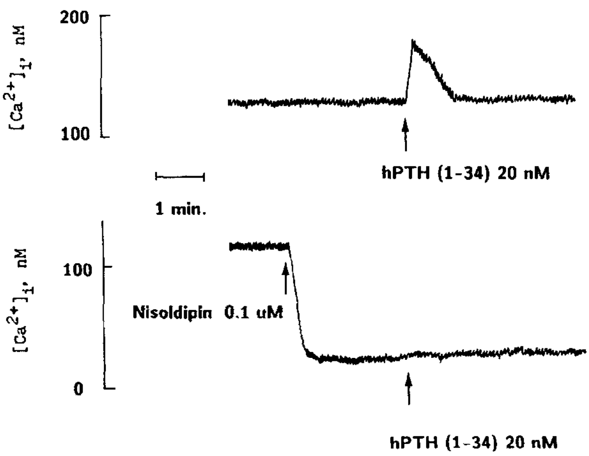


Figure 5. Effect of nisoldipin on the PTH action in vascular smooth muscle cells.

indicating that PTH activates calcium channels, thereby increasing $[Ca^{2+}]_i$.

Intracellular calcium concentration could also be elevated by inhibiting the activity of Na^+, K^+-ATPase. Inhibition of Na^+, K^+-ATPase increases intracellular Na^+ ion concentration which, in turn, stimulates the activity of the Na^+, Ca^{2+}-exchanger, thereby raising $[Ca^{2+}]_i$. This possibility appears to be unlikely, however, since 10^{-5} M ouabain did not change the effect of PTH, although addition of the drug resulted in a transient increase in $[Ca^{2+}]_i$, which was followed by a return to the control level (data not shown). The latter phenomenon may possibly result from the compensation due to a rise in Ca-ATPase activity.

The rise in $[Ca^{2+}]_i$ by PTH seems to be incompatible with the well-known vasodilating action of the hormone (4, 5). However, isoproterenol, which is a potent vasodilator, has recently been reported to produce a similar transient increase in $[Ca^{2+}]_i$ of tracheal smooth muscle (13). The action of isoproterenol has also been known to be mediated by cyclic AMP. The similarity in action of these two vasodilators strongly suggests that effect of PTH observed in the present study may be physiologically important, and that a transient increase in $[Ca^{2+}]_i$, per se, does not necessarily cause a contraction of vascular smooth muscle.

Our data also suggest that PTH may directly regulate cellular calcium homeostasis. In view of the previous data from our laboratory (14, 15) demonstrating that 1,25-dihydroxyvitamin D_3 directly regulates cellular calcium handling, present results provide further evidence that calcium regulating hormones directly regulate cellular calcium homeostasis, thereby regulating the cellular functions of vascular smooth muscle.

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